

Google Colab

```
Thu Feb 27 10:14:50 PM UTC 2025
env: YOSYS_EXE=yosys
env: OPENROAD_EXE=openroad
env: KLAYOUT_EXE=klayout
env: SOC25_STARTPACK=/content/soc25-starter-kit4
env: DESIGN_HOME=$(SOC25_STARTPACK)/uart16-demo
env: DESIGN_CONFIG=$(DESIGN_HOME)/config-soc25-uart-openroad-demo.mk
env: SYNTH_OPERATIONS_ARGS=""
/content/OpenROAD-flow-scripts/flow
mkdir -p ./results/nangate45/UART16/base ./logs/nangate45/UART16/base ./reports/nangate45/UART16/base ./objects/nangate45/UART16/base
env time -f 'Elapsed time: %E[h:]min:sec. CPU time: user %U sys %S (%P). Peak memory: %MKB.' yosys -v 3 -c /content/OpenROAD-flow-scripts/flow/scripts/synth_canonicalize.tcl -L log1.log
1. Executing Verilog-2005 frontend: /content/soc25-starter-kit4/uart16-demo/uart16.v
2. Executing Liberty frontend: ./objects/nangate45/UART16/base/lib/NangateOpenCellLibrary_typical.lib
3. Executing Liberty frontend: ./objects/nangate45/UART16/base/lib/NangateOpenCellLibrary_typical.lib
4. Executing Verilog-2005 frontend: /content/OpenROAD-flow-scripts/flow/platforms/nangate45/cells_clkgate.v
Using ABC area script.
Extracting clock period from SDC file: ./results/nangate45/UART16/base/clock_period.txt
Setting clock period to 0.46
5. Executing HIERARCHY pass (managing design hierarchy).
6. Executing AST frontend in derive mode using pre-parsed AST for module `UART16'.
6.1. Analyzing design hierarchy..
6.2. Executing AST frontend in derive mode using pre-parsed AST for module `PTOS'.
6.3. Executing AST frontend in derive mode using pre-parsed AST for module `STOP'.
6.4. Analyzing design hierarchy..
6.5. Analyzing design hierarchy..
7. Executing OPT_CLEAN pass (remove unused cells and wires).
Warning: Ignoring module PTOS because it contains processes (run 'proc' command first).
Warning: Ignoring module STOP because it contains processes (run 'proc' command first).
8. Executing RTLIL backend.
Warnings: 2 unique messages, 2 total
End of script. Logfile hash: b0292a5654, CPU: user 0.26s system 0.02s, MEM: 31.91 MB peak
Yosys 0.38+92 (git sha1 84116c9a3, x86_64-conda-linux-gnu-cc 11.2.0 -fvisibility-inlines-hidden -fmessage-length=0 -march=nocona -mtune=haswell -ftree-vectorize -fPIC -fstack-protector-strong -
Time spent: 94% 4x read_liberty (0 sec), 2% 1x hierarchy (0 sec), ...
Elapsed time: 0:00.40[h:]min:sec. CPU time: user 0.37 sys 0.03 (99%). Peak memory: 33168KB.
echo "Stupid" # @bash -c "for var in RCX_RULES DESIGN_NAME SDC_FILE IO_PLACER_H TIME_BIN GDS_FILES TIME_CMD GENERATE_ARTIFACTS_ON_FAILURE TAP_CELL_NAME CTS_CLUS
Stupid
cp /content/OpenROAD-flow-scripts/flow/UART16/base/* ./results/nangate45/UART16/base
echo "Silly"
Silly
# @bash -c "for var in RCX_RULES DESIGN_NAME SDC_FILE IO_PLACER_H TIME_BIN GDS_FILES TIME_CMD GENERATE_ARTIFACTS_ON_FAILURE TAP_CELL_NAME CTS_CLUSTER_DIAMETER
Running floorplan.tcl, stage 2_1_floorplan
[INFO ORD-0030] Using 2 thread(s).
[INFO ODB-0222] Reading LEF file: /content/OpenROAD-flow-scripts/flow/platforms/nangate45/lef/NangateOpenCellLibrary.tech.lef
[INFO ODB-0223] Created 22 technology layers
[INFO ODB-0224] Created 27 technology vias
[INFO ODB-0226] Finished LEF file: /content/OpenROAD-flow-scripts/flow/platforms/nangate45/lef/NangateOpenCellLibrary.tech.lef
[INFO ODB-0222] Reading LEF file: /content/OpenROAD-flow-scripts/flow/platforms/nangate45/lef/NangateOpenCellLibrary.macro.mod.lef
[INFO ODB-0225] Created 135 library cells
[INFO ODB-0226] Finished LEF file: /content/OpenROAD-flow-scripts/flow/platforms/nangate45/lef/NangateOpenCellLibrary.macro.mod.lef
[WARNING STA-0347] current_design for other than top cell not supported.

=====
Floorplan check_setup
=====
Warning: There are 41 unlocked register/latch pins.
Warning: There are 50 unconstrained endpoints.
number instances in verilog is 287
[WARNING IFP-0028] Core area lower left (1.000, 1.000) snapped to (1.140, 1.400).
[INFO IFP-0001] Added 25 rows of 186 site FreePDK45_38x28_10R_NP_162NW_340.
repair_timing -verbose -setup_margin 0 -repair_tns 100
[WARNING RSZ-0021] no estimated parasitics. Using wire load models.
[INFO RSZ-0033] No hold violations found.
Default units for flow
time 1ns
capacitance 1ff
resistance 1kohm
voltage 1v
current 1mA
power 1mW
distance 1um
Report metrics stage 2, floorplan final...

=====
floorplan final report_design_area
=====
Design area 509 u^2 41% utilization.
Elapsed time: 0:01.05[h:]min:sec. CPU time: user 0.85 sys 0.07 (87%). Peak memory: 88356KB.
Running io_placement_random.tcl, stage 2_2_floorplan_io
[INFO ORD-0030] Using 2 thread(s).
place_pins -hor_layers metal5 -ver_layers metal6 -random
Found 0 macro blocks.
Using 2 tracks default min distance between IO pins.
[INFO PPL-0007] Random pin placement.
Elapsed time: 0:00.81[h:]min:sec. CPU time: user 0.70 sys 0.06 (95%). Peak memory: 85948KB.
Running macro_place.tcl, stage 2_3_floorplan_macro
[INFO ORD-0030] Using 2 thread(s).
No macros found: Skipping macro_placement
Elapsed time: 0:00.50[h:]min:sec. CPU time: user 0.45 sys 0.04 (99%). Peak memory: 85644KB.
Running tapcell.tcl, stage 2_4_floorplan_tapcell
[INFO ORD-0030] Using 2 thread(s).
[INFO TAP-0004] Inserted 50 endcaps.
[INFO TAP-0005] Inserted 0 tapcells.
Elapsed time: 0:00.49[h:]min:sec. CPU time: user 0.45 sys 0.04 (99%). Peak memory: 85576KB.
Running pdn.tcl, stage 2_5_floorplan_pdn
[INFO ORD-0030] Using 2 thread(s).
[INFO PDN-0001] Inserting grid: grid
Elapsed time: 0:00.51[h:]min:sec. CPU time: user 0.45 sys 0.05 (99%). Peak memory: 88408KB.
cp ./results/nangate45/UART16/base/2_5_floorplan_pdn.odb ./results/nangate45/UART16/base/2_floorplan.odb
cp ./results/nangate45/UART16/base/2_1_floorplan.sdc ./results/nangate45/UART16/base/2_floorplan.sdc
Running global_place_skip_io.tcl, stage 3_1_place_gp_skip_io
[INFO ORD-0030] Using 2 thread(s).
[INFO GPL-0002] DBU: 2000
[INFO GPL-0003] SiteSize: 380 2800
[INFO GPL-0004] CoreAreaLxLy: 2280 2800
[INFO GPL-0005] CoreAreaUxUy: 72960 72800
[INFO GPL-0006] NumInstances: 337
[INFO GPL-0007] NumPlaceInstances: 287
[INFO GPL-0008] NumFixedInstances: 50
[INFO GPL-0009] NumDummyInstances: 0
[INFO GPL-0010] NumNets: 380
[INFO GPL-0011] NumPins: 975
[INFO GPL-0012] DieAreaLxLy: 0 0
[INFO GPL-0013] DieAreaUxUy: 75320 75320
[INFO GPL-0014] CoreAreaLxLy: 2280 2800
[INFO GPL-0015] CoreAreaUxUy: 72960 72800
[INFO GPL-0016] CoreArea: 4947600000
[INFO GPL-0017] NonPlaceInstsArea: 53200000
```

```

[INFO GPL-0018] PlaceInstsArea: 2034368000
[INFO GPL-0019] Util(%): 41.57
[INFO GPL-0020] StdInstsArea: 2034368000
[INFO GPL-0021] MacroInstsArea: 0
[INFO GPL-0031] FillerInit: NumGCells: 712
[INFO GPL-0032] FillerInit: NumGNets: 380
[INFO GPL-0033] FillerInit: NumGPins: 975
[INFO GPL-0023] TargetDensity: 1.00
[INFO GPL-0024] AveragePlaceInstArea: 7088390
[INFO GPL-0025] IdealBinArea: 7088390
[INFO GPL-0026] IdealBinCnt: 697
[INFO GPL-0027] TotalBinArea: 4947600000
[INFO GPL-0028] BinCnt: 16 16
[INFO GPL-0029] BinSize: 4418 4375
[INFO GPL-0030] NumBins: 256
Placement density is 0.5425217485427857, computed from PLACE_DENSITY_LB_ADDON 0.20 and lower bound 0.41565218567848206
global_placement -skip_io -density 0.5425217485427857 -pad_left 0 -pad_right 0
[INFO GPL-0002] DBU: 2000
[INFO GPL-0003] SiteSize: 380 2800
[INFO GPL-0004] CoreAreaLxLy: 2280 2800
[INFO GPL-0005] CoreAreaUxUy: 72960 72800
[INFO GPL-0006] NumInstances: 337
[INFO GPL-0007] NumPlaceInstances: 287
[INFO GPL-0008] NumFixedInstances: 50
[INFO GPL-0009] NumDummyInstances: 0
[INFO GPL-0010] NumNets: 380
[INFO GPL-0011] NumPins: 934
[INFO GPL-0012] DieAreaLxLy: 0 0
[INFO GPL-0013] DieAreaUxUy: 75320 75320
[INFO GPL-0014] CoreAreaLxLy: 2280 2800
[INFO GPL-0015] CoreAreaUxUy: 72960 72800
[INFO GPL-0016] CoreArea: 4947600000
[INFO GPL-0017] NonPlaceInstsArea: 53200000
[INFO GPL-0018] PlaceInstsArea: 2034368000
[INFO GPL-0019] Util(%): 41.57
[INFO GPL-0020] StdInstsArea: 2034368000
[INFO GPL-0021] MacroInstsArea: 0
[INFO GPL-0031] FillerInit: NumGCells: 379
[INFO GPL-0032] FillerInit: NumGNets: 380
[INFO GPL-0033] FillerInit: NumGPins: 934
[INFO GPL-0023] TargetDensity: 0.54
[INFO GPL-0024] AveragePlaceInstArea: 7088390
[INFO GPL-0025] IdealBinArea: 13065632
[INFO GPL-0026] IdealBinCnt: 378
[INFO GPL-0027] TotalBinArea: 4947600000
[INFO GPL-0028] BinCnt: 16 16
[INFO GPL-0029] BinSize: 4418 4375
[INFO GPL-0030] NumBins: 256
[NesterovSolve] Iter: 1 overflow: 0.972634 HPWL: 658226
[NesterovSolve] Iter: 10 overflow: 0.961878 HPWL: 433863
[NesterovSolve] Iter: 20 overflow: 0.95842 HPWL: 429162
[NesterovSolve] Iter: 30 overflow: 0.95842 HPWL: 427079
[NesterovSolve] Iter: 40 overflow: 0.958348 HPWL: 426618
[NesterovSolve] Iter: 50 overflow: 0.958088 HPWL: 426419
[NesterovSolve] Iter: 60 overflow: 0.957585 HPWL: 426922
[NesterovSolve] Iter: 70 overflow: 0.957135 HPWL: 428360
[NesterovSolve] Iter: 80 overflow: 0.956732 HPWL: 430660
[NesterovSolve] Iter: 90 overflow: 0.956088 HPWL: 435357
[NesterovSolve] Iter: 100 overflow: 0.954731 HPWL: 444128
[NesterovSolve] Iter: 110 overflow: 0.947612 HPWL: 459150
[NesterovSolve] Iter: 120 overflow: 0.936168 HPWL: 479524
[NesterovSolve] Iter: 130 overflow: 0.924823 HPWL: 504068
[NesterovSolve] Iter: 140 overflow: 0.894923 HPWL: 532174
[NesterovSolve] Iter: 150 overflow: 0.882692 HPWL: 561621
[NesterovSolve] Iter: 160 overflow: 0.865968 HPWL: 600915
[NesterovSolve] Iter: 170 overflow: 0.831034 HPWL: 664100
[NesterovSolve] Iter: 180 overflow: 0.755144 HPWL: 732213
[NesterovSolve] Iter: 190 overflow: 0.752231 HPWL: 795214
[NesterovSolve] Iter: 200 overflow: 0.703597 HPWL: 917989
[NesterovSolve] Iter: 210 overflow: 0.662449 HPWL: 987016
[NesterovSolve] Iter: 220 overflow: 0.625718 HPWL: 1064695
[NesterovSolve] Iter: 230 overflow: 0.598539 HPWL: 1158098
[NesterovSolve] Iter: 240 overflow: 0.550601 HPWL: 1239977
[NesterovSolve] Iter: 250 overflow: 0.509286 HPWL: 1315903
[NesterovSolve] Iter: 260 overflow: 0.46845 HPWL: 1380218
[NesterovSolve] Iter: 270 overflow: 0.422431 HPWL: 1468768
[NesterovSolve] Iter: 280 overflow: 0.378116 HPWL: 1551510
[NesterovSolve] Iter: 290 overflow: 0.333872 HPWL: 1612019
[NesterovSolve] Iter: 300 overflow: 0.295172 HPWL: 1643447
[NesterovSolve] Iter: 310 overflow: 0.266669 HPWL: 1680321
[NesterovSolve] Iter: 320 overflow: 0.233721 HPWL: 1720463
[NesterovSolve] Iter: 330 overflow: 0.20307 HPWL: 1742521
[NesterovSolve] Iter: 340 overflow: 0.175588 HPWL: 1790478
[NesterovSolve] Iter: 350 overflow: 0.146351 HPWL: 1813654
[NesterovSolve] Iter: 360 overflow: 0.122699 HPWL: 1840910
[NesterovSolve] Iter: 370 overflow: 0.104441 HPWL: 1864570
[NesterovSolve] Finished with Overflow: 0.097802
Elapsed time: 0:00.61[h]:min:sec. CPU time: user 0.65 sys 0.05 (114%). Peak memory: 86868KB.
Running io_placement.tcl, stage 3_2_place_iop
[INFO ORD-0030] Using 2 thread(s).
place_pins -hor_layers metal5 -ver_layers metal6
Found 0 macro blocks.
Using 2 tracks default min distance between IO pins.
[INFO PPL-0010] Tentative 0 to set up sections.
[INFO PPL-0001] Number of slots 262
[INFO PPL-0002] Number of I/O 41
[INFO PPL-0003] Number of I/O w/sink 31
[INFO PPL-0004] Number of I/O w/o sink 10
[INFO PPL-0005] Slots per section 200
[INFO PPL-0006] Slots increase factor 0.01
[INFO PPL-0008] Successfully assigned pins to sections.
[INFO PPL-0012] I/O nets HPWL: 301.20 um.
Elapsed time: 0:00.50[h]:min:sec. CPU time: user 0.45 sys 0.04 (99%). Peak memory: 86728KB.
Running global_place.tcl, stage 3_3_place_gp
[INFO ORD-0030] Using 2 thread(s).
[INFO GPL-0002] DBU: 2000
[INFO GPL-0003] SiteSize: 380 2800
[INFO GPL-0004] CoreAreaLxLy: 2280 2800
[INFO GPL-0005] CoreAreaUxUy: 72960 72800
[INFO GPL-0006] NumInstances: 337
[INFO GPL-0007] NumPlaceInstances: 287
[INFO GPL-0008] NumFixedInstances: 50
[INFO GPL-0009] NumDummyInstances: 0
[INFO GPL-0010] NumNets: 380
[INFO GPL-0011] NumPins: 975
[INFO GPL-0012] DieAreaLxLy: 0 0
[INFO GPL-0013] DieAreaUxUy: 75320 75320
[INFO GPL-0014] CoreAreaLxLy: 2280 2800
[INFO GPL-0015] CoreAreaUxUy: 72960 72800
[INFO GPL-0016] CoreArea: 4947600000
[INFO GPL-0017] NonPlaceInstsArea: 53200000
[INFO GPL-0018] PlaceInstsArea: 2034368000
[INFO GPL-0019] Util(%): 41.57

```

```
[INFO GPL-0020] StdInstsArea: 2034368000
[INFO GPL-0021] MacroInstsArea: 0
[INFO GPL-0031] FillerInit: NumGCells: 712
[INFO GPL-0032] FillerInit: NumGNets: 380
[INFO GPL-0033] FillerInit: NumGPins: 975
[INFO GPL-0023] TargetDensity: 1.00
[INFO GPL-0024] AveragePlaceInstArea: 7088390
[INFO GPL-0025] IdealBinArea: 7088390
[INFO GPL-0026] IdealBinCnt: 697
[INFO GPL-0027] TotalBinArea: 4947600000
[INFO GPL-0028] BinCnt: 16 16
[INFO GPL-0029] BinSize: 4418 4375
[INFO GPL-0030] NumBins: 256
Placement density is 0.5425217485427857, computed from PLACE_DENSITY_LB_ADDON 0.20 and lower bound 0.41565218567848206
global_placement -density 0.5425217485427857 -pad_left 0 -pad_right 0 -routability_driven -timing_driven
[INFO GPL-0002] DBU: 2000
[INFO GPL-0003] SiteSize: 380 2800
[INFO GPL-0004] CoreAreaLxLy: 2280 2800
[INFO GPL-0005] CoreAreaUxUy: 72960 72800
[INFO GPL-0006] NumInstances: 337
[INFO GPL-0007] NumPlaceInstances: 287
[INFO GPL-0008] NumFixedInstances: 50
[INFO GPL-0009] NumDummyInstances: 0
[INFO GPL-0010] NumNets: 380
[INFO GPL-0011] NumPins: 975
[INFO GPL-0012] DieAreaLxLy: 0 0
[INFO GPL-0013] DieAreaUxUy: 75320 75320
[INFO GPL-0014] CoreAreaLxLy: 2280 2800
[INFO GPL-0015] CoreAreaUxUy: 72960 72800
[INFO GPL-0016] CoreArea: 4947600000
[INFO GPL-0017] NonPlaceInstsArea: 53200000
[INFO GPL-0018] PlaceInstsArea: 2034368000
[INFO GPL-0019] Util(%): 41.57
[INFO GPL-0020] StdInstsArea: 2034368000
[INFO GPL-0021] MacroInstsArea: 0
[InitialPlace] Iter: 1 CG residual: 0.00000008 HPWL: 4637321
[InitialPlace] Iter: 2 CG residual: 0.00000009 HPWL: 3536346
[InitialPlace] Iter: 3 CG residual: 0.00000009 HPWL: 3466193
[InitialPlace] Iter: 4 CG residual: 0.00000009 HPWL: 3404858
[InitialPlace] Iter: 5 CG residual: 0.00000011 HPWL: 3367941
[INFO GPL-0031] FillerInit: NumGCells: 379
[INFO GPL-0032] FillerInit: NumGNets: 380
[INFO GPL-0033] FillerInit: NumGPins: 975
[INFO GPL-0023] TargetDensity: 0.54
[INFO GPL-0024] AveragePlaceInstArea: 7088390
[INFO GPL-0025] IdealBinArea: 13065632
[INFO GPL-0026] IdealBinCnt: 378
[INFO GPL-0027] TotalBinArea: 4947600000
[INFO GPL-0028] BinCnt: 16 16
[INFO GPL-0029] BinSize: 4418 4375
[INFO GPL-0030] NumBins: 256
[NesterovSolve] Iter: 1 overflow: 0.83495 HPWL: 1294141
[NesterovSolve] Iter: 10 overflow: 0.800922 HPWL: 1316907
[INFO GPL-0100] worst slack 1.81e-10
[INFO GPL-0103] Weighted 36 nets.
[NesterovSolve] Iter: 20 overflow: 0.782467 HPWL: 1317063
[NesterovSolve] Iter: 30 overflow: 0.771365 HPWL: 1323739
[NesterovSolve] Iter: 40 overflow: 0.766441 HPWL: 1326988
[NesterovSolve] Iter: 50 overflow: 0.768462 HPWL: 1332458
[NesterovSolve] Iter: 60 overflow: 0.761835 HPWL: 1337680
[NesterovSolve] Iter: 70 overflow: 0.760113 HPWL: 1342120
[NesterovSolve] Iter: 80 overflow: 0.759515 HPWL: 1345035
[NesterovSolve] Iter: 90 overflow: 0.75756 HPWL: 1345872
[NesterovSolve] Iter: 100 overflow: 0.755689 HPWL: 1345692
[NesterovSolve] Iter: 110 overflow: 0.753774 HPWL: 1345601
[NesterovSolve] Iter: 120 overflow: 0.75113 HPWL: 1348862
[NesterovSolve] Iter: 130 overflow: 0.746399 HPWL: 1356370
[NesterovSolve] Iter: 140 overflow: 0.741236 HPWL: 1369463
[NesterovSolve] Iter: 150 overflow: 0.716543 HPWL: 1389292
[NesterovSolve] Iter: 160 overflow: 0.704813 HPWL: 1416705
[NesterovSolve] Iter: 170 overflow: 0.690062 HPWL: 1450275
[NesterovSolve] Iter: 180 overflow: 0.659096 HPWL: 1492253
[NesterovSolve] Iter: 190 overflow: 0.642692 HPWL: 1533982
[INFO GPL-0100] worst slack 1.8e-10
[INFO GPL-0103] Weighted 35 nets.
[NesterovSolve] Iter: 200 overflow: 0.605199 HPWL: 1570377
[NesterovSolve] Snapshot saved at iter = 203
[NesterovSolve] Iter: 210 overflow: 0.584007 HPWL: 1610239
[NesterovSolve] Iter: 220 overflow: 0.556945 HPWL: 1652022
[NesterovSolve] Iter: 230 overflow: 0.517268 HPWL: 1701543
[NesterovSolve] Iter: 240 overflow: 0.483939 HPWL: 1736787
[INFO GPL-0100] worst slack 1.8e-10
[INFO GPL-0103] Weighted 36 nets.
[NesterovSolve] Iter: 250 overflow: 0.44775 HPWL: 1787039
[NesterovSolve] Iter: 260 overflow: 0.408281 HPWL: 1828770
[NesterovSolve] Iter: 270 overflow: 0.368293 HPWL: 1871882
[NesterovSolve] Iter: 280 overflow: 0.33001 HPWL: 1912377
[NesterovSolve] Iter: 290 overflow: 0.301252 HPWL: 1947044
[INFO GPL-0100] worst slack 1.8e-10
[INFO GPL-0103] Weighted 36 nets.
[NesterovSolve] Iter: 300 overflow: 0.271251 HPWL: 1988336
[NesterovSolve] Iter: 310 overflow: 0.245323 HPWL: 2024858
[NesterovSolve] Iter: 320 overflow: 0.214779 HPWL: 2064824
[INFO GPL-0100] worst slack 1.8e-10
[INFO GPL-0103] Weighted 34 nets.
[INFO GPL-0075] Routability numCall: 1 inflationIterCnt: 1 bloatIterCnt: 0
[INFO GPL-0036] TileLxLy: 0 0
[INFO GPL-0037] TileSize: 4200 4200
[INFO GPL-0038] TileCnt: 17 17
[INFO GPL-0039] numRoutingLayers: 10
[INFO GPL-0040] NumTiles: 289
[INFO GPL-0063] TotalRouteOverflowH2: 0
[INFO GPL-0064] TotalRouteOverflowV2: 0
[INFO GPL-0065] OverflowTileCnt2: 0
[INFO GPL-0066] 0.5%RC: 0.9318182021379471
[INFO GPL-0067] 1.0%RC: 0.9116883317629496
[INFO GPL-0068] 2.0%RC: 0.8665472536251463
[INFO GPL-0069] 5.0%RC: 0.798520935078462
[INFO GPL-0070] 0.5rcK: 1
[INFO GPL-0071] 1.0rcK: 1
[INFO GPL-0072] 2.0rcK: 0
[INFO GPL-0073] 5.0rcK: 0
[INFO GPL-0074] FinalRC: 0.9217533
[NesterovSolve] Iter: 330 overflow: 0.18504 HPWL: 2095671
[NesterovSolve] Iter: 340 overflow: 0.158281 HPWL: 2132834
[INFO GPL-0100] worst slack 1.8e-10
[INFO GPL-0103] Weighted 34 nets.
[NesterovSolve] Iter: 350 overflow: 0.136002 HPWL: 2160269
[NesterovSolve] Iter: 360 overflow: 0.108415 HPWL: 2179663
[NesterovSolve] Finished with Overflow: 0.099347
Report metrics stage 3, global place...
```

=====

```
global place report_design_area
-----
Design area 522 u^2 42% utilization.
Elapsed time: 0:01.31[h:]min:sec. CPU time: user 1.41 sys 0.12 (117%). Peak memory: 188556KB.
Running resize.tcl, stage 3_4_place_resized
[INFO ORD-0030] Using 2 thread(s).
Perform port buffering...
[INFO RSZ-0027] Inserted 13 input buffers.
[INFO RSZ-0028] Inserted 9 output buffers.
Perform buffer insertion and gate resizing...
repair_design -verbose
[INFO RSZ-0058] Using max wire length 634um.
Iteration | Resized | Buffers | Nets repaired | Remaining
-----
  0 | 0 | 0 | 0 | 402
 10 | 2 | 0 | 0 | 392
 20 | 4 | 0 | 0 | 382
 30 | 4 | 0 | 0 | 372
 40 | 10 | 0 | 0 | 362
 50 | 10 | 0 | 0 | 352
 60 | 12 | 0 | 0 | 342
 70 | 12 | 0 | 0 | 332
 80 | 12 | 0 | 0 | 322
 90 | 12 | 0 | 0 | 312
100 | 12 | 0 | 0 | 302
110 | 12 | 0 | 0 | 292
120 | 12 | 0 | 0 | 282
130 | 14 | 0 | 0 | 272
140 | 14 | 0 | 0 | 262
150 | 15 | 0 | 0 | 252
160 | 15 | 0 | 0 | 242
170 | 15 | 0 | 0 | 232
180 | 16 | 0 | 0 | 222
190 | 16 | 0 | 0 | 212
200 | 16 | 0 | 0 | 202
210 | 16 | 0 | 0 | 192
220 | 16 | 0 | 0 | 182
230 | 16 | 0 | 0 | 172
240 | 16 | 0 | 0 | 162
250 | 17 | 0 | 0 | 152
260 | 18 | 0 | 0 | 142
270 | 18 | 0 | 0 | 132
280 | 18 | 0 | 0 | 122
290 | 18 | 0 | 0 | 112
300 | 18 | 0 | 0 | 102
310 | 18 | 0 | 0 | 92
320 | 18 | 0 | 0 | 82
330 | 18 | 0 | 0 | 72
340 | 18 | 0 | 0 | 62
350 | 18 | 0 | 0 | 52
360 | 20 | 0 | 0 | 42
370 | 21 | 0 | 0 | 32
380 | 21 | 0 | 0 | 22
390 | 21 | 0 | 0 | 12
400 | 21 | 0 | 0 | 2
final | 21 | 0 | 0 | 0
-----
[INFO RSZ-0039] Resized 21 instances.
Repair tie lo fanout...
[INFO RSZ-0042] Inserted 8 tie LOGIC0_X1 instances.
Repair tie hi fanout...
Floating nets:
Report metrics stage 3, resizer...

=====
resizer report_design_area
-----
Design area 537 u^2 43% utilization.
Instance count before 337, after 366
Pin count before 934, after 985
Elapsed time: 0:00.78[h:]min:sec. CPU time: user 0.71 sys 0.06 (99%). Peak memory: 94256KB.
Running detail_place.tcl, stage 3_5_place_dp
[INFO ORD-0030] Using 2 thread(s).
Placement Analysis
-----
total displacement      308.9 u
average displacement    0.8 u
max displacement        2.7 u
original HPWL           1102.1 u
legalized HPWL          1358.0 u
delta HPWL              23 %

Detailed placement improvement.
Importing netlist into detailed improver.
[INFO DPO-0100] Creating network with 366 cells, 41 terminals, 409 edges and 1026 pins.
[INFO DPO-0109] Network stats: inst 407, edges 409, pins 1026
[INFO DPO-0110] Number of regions is 1
[INFO DPO-0401] Setting random seed to 1.
[INFO DPO-0402] Setting maximum displacement 5 1 to 14000 2800 units.
[INFO DPO-0320] Collected 91 fixed cells (excluded terminal_NI).
[INFO DPO-0318] Collected 316 single height cells.
[INFO DPO-0321] Collected 0 wide cells.
[INFO DPO-0322] Image (2280, 2800) - (72960, 72800)
[INFO DPO-0310] Assigned 316 cells into segments. Movement in X-direction is 0.000000, movement in Y-direction is 0.000000.
[INFO DPO-0313] Found 0 cells in wrong regions.
[INFO DPO-0315] Found 0 row alignment problems.
[INFO DPO-0314] Found 0 site alignment problems.
[INFO DPO-0311] Found 0 overlaps between adjacent cells.
[INFO DPO-0312] Found 0 edge spacing violations and 0 padding violations.
[INFO DPO-0303] Running algorithm for independent set matching.
[INFO DPO-0300] Set matching objective is wirelength.
[INFO DPO-0301] Pass 1 of matching; objective is 2.782205e+06.
[INFO DPO-0302] End of matching; objective is 2.782205e+06, improvement is 0.00 percent.
[INFO DPO-0303] Running algorithm for global swaps.
[INFO DPO-0306] Pass 1 of global swaps; hpwl is 2.654725e+06.
[INFO DPO-0306] Pass 2 of global swaps; hpwl is 2.636785e+06.
[INFO DPO-0307] End of global swaps; objective is 2.636785e+06, improvement is 5.23 percent.
[INFO DPO-0303] Running algorithm for vertical swaps.
[INFO DPO-0308] Pass 1 of vertical swaps; hpwl is 2.626195e+06.
[INFO DPO-0309] End of vertical swaps; objective is 2.626195e+06, improvement is 0.40 percent.
[INFO DPO-0303] Running algorithm for reordering.
[INFO DPO-0304] Pass 1 of reordering; objective is 2.605635e+06.
[INFO DPO-0305] End of reordering; objective is 2.605635e+06, improvement is 0.78 percent.
[INFO DPO-0303] Running algorithm for random improvement.
[INFO DPO-0324] Random improver is using displacement generator.
[INFO DPO-0325] Random improver is using hpwl objective.
[INFO DPO-0326] Random improver cost string is (a).
[INFO DPO-0332] End of pass, Generator displacement called 6320 times.
[INFO DPO-0335] Generator displacement, Cumulative attempts 6320, swaps 1006, moves 2254 since last reset.
[INFO DPO-0333] End of pass, Objective hpwl, Initial cost 2.605635e+06, Scratch cost 2.556640e+06, Incremental cost 2.556640e+06, Mismatch? N
[INFO DPO-0338] End of pass, Total cost is 2.556640e+06.
[INFO DPO-0327] Pass 1 of random improver; improvement in cost is 1.88 percent.
[INFO DPO-0332] End of pass, Generator displacement called 6320 times.
```

```
[INFO DPO-0335] Generator displacement, Cumulative attempts 12640, swaps 2039, moves 4505 since last reset.
[INFO DPO-0333] End of pass, Objective hpwl, Initial cost 2.556640e+06, Scratch cost 2.533080e+06, Incremental cost 2.533080e+06, Mismatch? N
[INFO DPO-0338] End of pass, Total cost is 2.533080e+06.
[INFO DPO-0327] Pass 2 of random improver; improvement in cost is 0.92 percent.
[INFO DPO-0328] End of random improver; improvement is 2.784542 percent.
[INFO DPO-0380] Cell flipping.
[INFO DPO-0382] Changed 141 cell orientations for row compatibility.
[INFO DPO-0383] Performed 104 cell flips.
[INFO DPO-0384] End of flipping; objective is 2.468040e+06, improvement is 2.57 percent.
[INFO DPO-0313] Found 0 cells in wrong regions.
[INFO DPO-0315] Found 0 row alignment problems.
[INFO DPO-0314] Found 0 site alignment problems.
[INFO DPO-0311] Found 0 overlaps between adjacent cells.
[INFO DPO-0312] Found 0 edge spacing violations and 0 padding violations.
Detailed Improvement Results
```

```
-----
Original HPWL      1358.0 u
Final HPWL         1187.5 u
Delta HPWL         -12.6 %
```

```
[INFO DPL-0020] Mirrored 7 instances
[INFO DPL-0021] HPWL before      1187.5 u
[INFO DPL-0022] HPWL after      1187.5 u
[INFO DPL-0023] HPWL delta      -0.0 %
[INFO FLW-0012] Placement violations .
Report metrics stage 3, detailed place...
```

```
=====
detailed place report_design_area
-----
```

```
Design area 537 u^2 43% utilization.
Elapsed time: 0:00.76[h]:min:sec. CPU time: user 0.70 sys 0.05 (99%). Peak memory: 89264KB.
cp ./results/nangate45/UART16/base/3_5_place_dp.odb ./results/nangate45/UART16/base/3_place.odb
cp ./results/nangate45/UART16/base/2_floorplan.sdc ./results/nangate45/UART16/base/3_place.sdc
Running cts.tcl, stage 4_1.cts
[INFO ORD-0030] Using 2 thread(s).
clock_tree_synthesis -sink_clustering_enable -balance_levels
[INFO CTS-0050] Root buffer is CLKBUF_X3.
[INFO CTS-0051] Sink buffer is CLKBUF_X3.
[INFO CTS-0052] The following clock buffers will be used for CTS:
                CLKBUF_X3
[INFO CTS-0049] Characterization buffer is CLKBUF_X3.
[INFO CTS-0007] Net "clk" found for clock "core_clock".
[INFO CTS-0010] Clock net "clk" has 18 sinks.
[INFO CTS-0008] TritonCTS found 1 clock nets.
[INFO CTS-0097] Characterization used 1 buffer(s) types.
[INFO CTS-0200] 0 placement blockages have been identified.
[INFO CTS-0201] 0 placed hard macros will be treated like blockages.
[INFO CTS-0027] Generating H-Tree topology for net clk.
[INFO CTS-0028] Total number of sinks: 18.
[INFO CTS-0090] Sinks will be clustered based on buffer max cap.
[INFO CTS-0030] Number of static layers: 0.
[INFO CTS-0020] Wire segment unit: 14000 dbu (7 um).
[INFO CTS-0023] Original sink region: [(11970, 46370), (69350, 68770)].
[INFO CTS-0024] Normalized sink region: [(0.855, 3.31214), (4.95357, 4.91214)].
[INFO CTS-0025] Width: 4.0986.
[INFO CTS-0026] Height: 1.6000.
Level 1
Direction: Horizontal
Sinks per sub-region: 9
Sub-region size: 2.0493 X 1.6000
[INFO CTS-0034] Segment length (rounded): 1.
[INFO CTS-0032] Stop criterion found. Max number of sinks is 15.
[INFO CTS-0035] Number of sinks covered: 18.
[INFO CTS-0018] Created 3 clock buffers.
[INFO CTS-0012] Minimum number of buffers in the clock path: 2.
[INFO CTS-0013] Maximum number of buffers in the clock path: 2.
[INFO CTS-0015] Created 3 clock nets.
[INFO CTS-0016] Fanout distribution for the current clock = 9:2..
[INFO CTS-0017] Max level of the clock tree: 1.
[INFO CTS-0098] Clock net "clk"
[INFO CTS-0099] Sinks 18
[INFO CTS-0100] Leaf buffers 0
[INFO CTS-0101] Average sink wire length 35.09 um
[INFO CTS-0102] Path depth 2 - 2
[INFO RSZ-0058] Using max wire length 634um.
```

```
Placement Analysis
-----
total displacement      4.7 u
average displacement    0.0 u
max displacement        2.0 u
original HPWL           1226.8 u
legalized HPWL          1274.0 u
delta HPWL              4 %
```

```
repair_timing -verbose -setup_margin 0 -hold_margin 0 -repair_tns 100
[INFO RSZ-0033] No hold violations found.
Placement Analysis
```

```
-----
total displacement      0.0 u
average displacement    0.0 u
max displacement        0.0 u
original HPWL           1274.0 u
legalized HPWL          1274.0 u
delta HPWL              0 %
```

```
Report metrics stage 4, cts final...
```

```
=====
cts final report_design_area
-----
```

```
Design area 541 u^2 44% utilization.
Elapsed time: 0:03.04[h]:min:sec. CPU time: user 2.88 sys 0.07 (97%). Peak memory: 97856KB.
cp ./results/nangate45/UART16/base/4_1.cts.odb ./results/nangate45/UART16/base/4_cts.odb
Running global_route.tcl, stage 5_1.grt
[INFO ORD-0030] Using 2 thread(s).
[INFO DRT-0149] Reading tech and libs.
```

```
Units:                2000
Number of layers:      21
Number of macros:      135
Number of vias:        27
Number of viarulegen:   19
```

```
[INFO DRT-0150] Reading design.
```

```
Design:                UART16
Die area:              ( 0 0 ) ( 75320 75320 )
Number of track patterns: 20
Number of DEF vias:    0
Number of components:  369
Number of terminals:   41
Number of snets:       2
```

Number of nets: 412

[INFO DRT-0167] List of default vias:

Layer via2
default via: via2_5
Layer via3
default via: via3_2
Layer via4
default via: via4_0
Layer via5
default via: via5_0
Layer via6
default via: via6_0
Layer via7
default via: via7_0
Layer via8
default via: via8_0
Layer via9
default via: via9_0

[INFO DRT-0162] Library cell analysis.

[INFO DRT-0163] Instance analysis.

[INFO DRT-0164] Number of unique instances = 50.

[INFO DRT-0168] Init region query.

[INFO DRT-0024] Complete active.

[INFO DRT-0024] Complete Fr_VIA.

[INFO DRT-0024] Complete metal1.

[INFO DRT-0024] Complete via1.

[INFO DRT-0024] Complete metal2.

[INFO DRT-0024] Complete via2.

[INFO DRT-0024] Complete metal3.

[INFO DRT-0024] Complete via3.

[INFO DRT-0024] Complete metal4.

[INFO DRT-0024] Complete via4.

[INFO DRT-0024] Complete metal5.

[INFO DRT-0024] Complete via5.

[INFO DRT-0024] Complete metal6.

[INFO DRT-0024] Complete via6.

[INFO DRT-0024] Complete metal7.

[INFO DRT-0024] Complete via7.

[INFO DRT-0024] Complete metal8.

[INFO DRT-0024] Complete via8.

[INFO DRT-0024] Complete metal9.

[INFO DRT-0024] Complete via9.

[INFO DRT-0024] Complete metal10.

[INFO DRT-0033] active shape region query size = 0.

[INFO DRT-0033] FR_VIA shape region query size = 0.

[INFO DRT-0033] metal1 shape region query size = 5888.

[INFO DRT-0033] via1 shape region query size = 78.

[INFO DRT-0033] metal2 shape region query size = 52.

[INFO DRT-0033] via2 shape region query size = 78.

[INFO DRT-0033] metal3 shape region query size = 52.

[INFO DRT-0033] via3 shape region query size = 78.

[INFO DRT-0033] metal4 shape region query size = 28.

[INFO DRT-0033] via4 shape region query size = 0.

[INFO DRT-0033] metal5 shape region query size = 7.

[INFO DRT-0033] via5 shape region query size = 0.

[INFO DRT-0033] metal6 shape region query size = 34.

[INFO DRT-0033] via6 shape region query size = 0.

[INFO DRT-0033] metal7 shape region query size = 0.

[INFO DRT-0033] via7 shape region query size = 0.

[INFO DRT-0033] metal8 shape region query size = 0.

[INFO DRT-0033] via8 shape region query size = 0.

[INFO DRT-0033] metal9 shape region query size = 0.

[INFO DRT-0033] via9 shape region query size = 0.

[INFO DRT-0033] metal10 shape region query size = 0.

[INFO DRT-0165] Start pin access.

[INFO DRT-0076] Complete 100 pins.

[INFO DRT-0078] Complete 138 pins.

[INFO DRT-0081] Complete 46 unique inst patterns.

[INFO DRT-0084] Complete 179 groups.

#scanned instances = 369

#unique instances = 50

#stdCellGenAp = 1153

#stdCellValidPlanarAp = 0

#stdCellValidViaAp = 827

#stdCellPinNoAp = 0

#stdCellPinCnt = 991

#instTermValidViaApCnt = 0

#macroGenAp = 0

#macroValidPlanarAp = 0

#macroValidViaAp = 0

#macroNoAp = 0

[INFO DRT-0166] Complete pin access.

[INFO DRT-0267] cpu time = 00:00:07, elapsed time = 00:00:03, memory = 88.10 (MB), peak = 88.10 (MB)

global_route -congestion_report_file ./reports/nangate45/UART16/base/congestion.rpt -congestion_iterations 30 -congestion_report_iter_step 5 -verbose

[INFO GRT-0020] Min routing layer: metal2

[INFO GRT-0021] Max routing layer: metal10

[INFO GRT-0022] Global adjustment: 0%

[INFO GRT-0023] Grid origin: (0, 0)

[INFO GRT-0043] No OR_DEFAULT vias defined.

[INFO GRT-0088] Layer metal1 Track-Pitch = 0.1400 line-2-Via Pitch: 0.1350

[INFO GRT-0088] Layer metal2 Track-Pitch = 0.1900 line-2-Via Pitch: 0.1400

[INFO GRT-0088] Layer metal3 Track-Pitch = 0.1400 line-2-Via Pitch: 0.1400

[INFO GRT-0088] Layer metal4 Track-Pitch = 0.2800 line-2-Via Pitch: 0.2800

[INFO GRT-0088] Layer metal5 Track-Pitch = 0.2800 line-2-Via Pitch: 0.2800

[INFO GRT-0088] Layer metal6 Track-Pitch = 0.2800 line-2-Via Pitch: 0.2800

[INFO GRT-0088] Layer metal7 Track-Pitch = 0.8000 line-2-Via Pitch: 0.8000

[INFO GRT-0088] Layer metal8 Track-Pitch = 0.8000 line-2-Via Pitch: 0.8000

[INFO GRT-0088] Layer metal9 Track-Pitch = 1.6000 line-2-Via Pitch: 1.6000

[INFO GRT-0088] Layer metal10 Track-Pitch = 1.6000 line-2-Via Pitch: 1.6000

[INFO GRT-0019] Found 4 clock nets.

[INFO GRT-0001] Minimum degree: 2

[INFO GRT-0002] Maximum degree: 30

[INFO GRT-0003] Macros: 0

[INFO GRT-0004] Blockages: 0

[INFO GRT-0053] Routing resources analysis:

Layer	Direction	Routing	Original	Derated	Resource
		Resources	Resources	Resources	Reduction (%)
metal1	Horizontal	0	0	0	0.00%
metal2	Vertical	3349	1440	1440	57.00%
metal3	Horizontal	4573	2016	2016	55.92%
metal4	Vertical	2142	1378	1378	35.67%
metal5	Horizontal	2142	1434	1434	33.05%
metal6	Vertical	2142	1434	1434	33.05%
metal7	Horizontal	629	304	304	51.67%
metal8	Vertical	629	304	304	51.67%
metal9	Horizontal	306	256	256	16.34%
metal10	Vertical	306	256	256	16.34%

[INFO GRT-0197] Via related to pin nodes: 1464

[INFO GRT-0198] Via related Steiner nodes: 23
[INFO GRT-0199] Via filling finished.
[INFO GRT-0111] Final number of vias: 1755
[INFO GRT-0112] Final usage 3D: 5977

[INFO GRT-0096] Final congestion report:

Layer	Resource	Demand	Usage (%)	Max H	Max V	Total Overflow
metal1	0	0	0.00%	0	0	0
metal2	1440	353	24.51%	0	0	0
metal3	2016	339	16.82%	0	0	0
metal4	1378	14	1.02%	0	0	0
metal5	1434	6	0.42%	0	0	0
metal6	1434	0	0.00%	0	0	0
metal7	304	0	0.00%	0	0	0
metal8	304	0	0.00%	0	0	0
metal9	256	0	0.00%	0	0	0
metal10	256	0	0.00%	0	0	0
Total	8822	712	8.07%	0	0	0

[INFO GRT-0018] Total wirelength: 2732 um

[INFO GRT-0014] Routed nets: 347

Perform buffer insertion and gate resizing...

repair_design -verbose

[INFO RSZ-0058] Using max wire length 634um.

Iteration | Resized | Buffers | Nets repaired | Remaining

0	0	0	0	412
10	0	0	0	402
20	0	0	0	392
30	0	0	0	382
40	0	0	0	372
50	0	0	0	362
60	0	0	0	352
70	0	0	0	342
80	0	0	0	332
90	0	0	0	322
100	0	0	0	312
110	0	0	0	302
120	0	0	0	292
130	0	0	0	282
140	0	0	0	272
150	0	0	0	262
160	0	0	0	252
170	0	0	0	242
180	0	0	0	232
190	0	0	0	222
200	0	0	0	212
210	0	0	0	202
220	0	0	0	192
230	0	0	0	182
240	0	0	0	172
250	0	0	0	162
260	1	0	0	152
270	1	0	0	142
280	1	0	0	132
290	1	0	0	122
300	1	0	0	112
310	1	0	0	102
320	1	0	0	92
330	1	0	0	82
340	1	0	0	72
350	1	0	0	62
360	1	0	0	52
370	1	0	0	42
380	1	0	0	32
390	1	0	0	22
400	1	0	0	12
410	1	0	0	2
final	1	0	0	0

[INFO RSZ-0039] Resized 1 instances.

global_route -start_incremental

detailed_placement

Placement Analysis

total displacement	0.0 u
average displacement	0.0 u
max displacement	0.0 u
original HPWL	1274.3 u
legalized HPWL	1274.3 u
delta HPWL	0 %

global_route -end_incremental -congestion_report_file ./reports/nangate45/UART16/base/congestion_post_repair_design.rpt

Repair setup and hold violations...

repair_timing -verbose -setup_margin 0 -hold_margin 0 -repair_tns 100

[INFO RSZ-0033] No hold violations found.

global_route -start_incremental

detailed_placement

Placement Analysis

total displacement	0.0 u
average displacement	0.0 u
max displacement	0.0 u
original HPWL	1274.3 u
legalized HPWL	1274.3 u
delta HPWL	0 %

global_route -end_incremental -congestion_report_file ./reports/nangate45/UART16/base/congestion_post_repair_timing.rpt

global_route -start_incremental

global_route -end_incremental -congestion_report_file ./reports/nangate45/UART16/base/congestion_post_recover_power.rpt

Repair antennas...

[WARNING GRT-0246] No diode with LEF class CORE ANTENNACELL found.

[INFO ANT-0002] Found 0 net violations.

[INFO ANT-0001] Found 0 pin violations.

Estimate parasitics...

Report metrics stage 5, global route...

global route report_design_area

Design area 542 u^2 44% utilization.

Elapsed time: 0:05.15[h]:min:sec. CPU time: user 8.48 sys 0.15 (167%). Peak memory: 191496KB.

Running detail_route.tcl, stage 5_2_route

[INFO ORD-0030] Using 2 thread(s).

detailed_route -output_drc ./reports/nangate45/UART16/base/5_route_drc.rpt -output_maze ./results/nangate45/UART16/base/maze.log -bottom_routing_layer metal2 -top_routing_layer metal10 -droute_

[INFO DRT-0149] Reading tech and libs.

Units: 2000
Number of layers: 21
Number of macros: 135
Number of vias: 27

Number of viarulegen: 19

[INFO DRT-0150] Reading design.

Design: UART16
Die area: (0 0) (75320 75320)
Number of track patterns: 20
Number of DEF vias: 0
Number of components: 369
Number of terminals: 41
Number of snets: 2
Number of nets: 412

[INFO DRT-0167] List of default vias:

Layer via2
default via: via2_5
Layer via3
default via: via3_2
Layer via4
default via: via4_0
Layer via5
default via: via5_0
Layer via6
default via: via6_0
Layer via7
default via: via7_0
Layer via8
default via: via8_0
Layer via9
default via: via9_0

[INFO DRT-0162] Library cell analysis.

[INFO DRT-0163] Instance analysis.

[INFO DRT-0164] Number of unique instances = 50.

[INFO DRT-0168] Init region query.

[INFO DRT-0024] Complete active.
[INFO DRT-0024] Complete Fr_VIA.
[INFO DRT-0024] Complete metal1.
[INFO DRT-0024] Complete via1.
[INFO DRT-0024] Complete metal2.
[INFO DRT-0024] Complete via2.
[INFO DRT-0024] Complete metal3.
[INFO DRT-0024] Complete via3.
[INFO DRT-0024] Complete metal4.
[INFO DRT-0024] Complete via4.
[INFO DRT-0024] Complete metal5.
[INFO DRT-0024] Complete via5.
[INFO DRT-0024] Complete metal6.
[INFO DRT-0024] Complete via6.
[INFO DRT-0024] Complete metal7.
[INFO DRT-0024] Complete via7.
[INFO DRT-0024] Complete metal8.
[INFO DRT-0024] Complete via8.
[INFO DRT-0024] Complete metal9.
[INFO DRT-0024] Complete via9.
[INFO DRT-0024] Complete metal10.

[INFO DRT-0033] active shape region query size = 0.

[INFO DRT-0033] FR_VIA shape region query size = 0.

[INFO DRT-0033] metal1 shape region query size = 5894.

[INFO DRT-0033] via1 shape region query size = 78.

[INFO DRT-0033] metal2 shape region query size = 52.

[INFO DRT-0033] via2 shape region query size = 78.

[INFO DRT-0033] metal3 shape region query size = 52.

[INFO DRT-0033] via3 shape region query size = 78.

[INFO DRT-0033] metal4 shape region query size = 28.

[INFO DRT-0033] via4 shape region query size = 0.

[INFO DRT-0033] metal5 shape region query size = 7.

[INFO DRT-0033] via5 shape region query size = 0.

[INFO DRT-0033] metal6 shape region query size = 34.

[INFO DRT-0033] via6 shape region query size = 0.

[INFO DRT-0033] metal7 shape region query size = 0.

[INFO DRT-0033] via7 shape region query size = 0.

[INFO DRT-0033] metal8 shape region query size = 0.

[INFO DRT-0033] via8 shape region query size = 0.

[INFO DRT-0033] metal9 shape region query size = 0.

[INFO DRT-0033] via9 shape region query size = 0.

[INFO DRT-0033] metal10 shape region query size = 0.

[INFO DRT-0165] Start pin access.

[INFO DRT-0076] Complete 100 pins.

[INFO DRT-0078] Complete 138 pins.

[INFO DRT-0081] Complete 46 unique inst patterns.

[INFO DRT-0084] Complete 179 groups.

#scanned instances = 369

#unique instances = 50

#stdCellGenAp = 1153

#stdCellValidPlanarAp = 0

#stdCellValidViaAp = 827

#stdCellPinNoAp = 0

#stdCellPinCnt = 991

#instTermValidViaApCnt = 0

#macroGenAp = 0

#macroValidPlanarAp = 0

#macroValidViaAp = 0

#macroNoAp = 0

[INFO DRT-0166] Complete pin access.

[INFO DRT-0267] cpu time = 00:00:07, elapsed time = 00:00:03, memory = 97.30 (MB), peak = 97.30 (MB)

Number of guides: 2188

[INFO DRT-0169] Post process guides.

[INFO DRT-0176] GCELLGRID X 0 DO 17 STEP 4200 ;

[INFO DRT-0177] GCELLGRID Y 0 DO 17 STEP 4200 ;

[INFO DRT-0028] Complete active.

[INFO DRT-0028] Complete Fr_VIA.

[INFO DRT-0028] Complete metal1.

[INFO DRT-0028] Complete via1.

[INFO DRT-0028] Complete metal2.

[INFO DRT-0028] Complete via2.

[INFO DRT-0028] Complete metal3.

[INFO DRT-0028] Complete via3.

[INFO DRT-0028] Complete metal4.

[INFO DRT-0028] Complete via4.

[INFO DRT-0028] Complete metal5.

[INFO DRT-0028] Complete via5.

[INFO DRT-0028] Complete metal6.

[INFO DRT-0028] Complete via6.

[INFO DRT-0028] Complete metal7.

[INFO DRT-0028] Complete via7.

[INFO DRT-0028] Complete metal8.

[INFO DRT-0028] Complete via8.

[INFO DRT-0028] Complete metal9.

[INFO DRT-0028] Complete via9.

[INFO DRT-0028] Complete metal10.

[INFO DRT-0178] Init guide query.

```

[INFO DRT-0035] Complete active (guide).
[INFO DRT-0035] Complete Fr_VIA (guide).
[INFO DRT-0035] Complete metal1 (guide).
[INFO DRT-0035] Complete via1 (guide).
[INFO DRT-0035] Complete metal2 (guide).
[INFO DRT-0035] Complete via2 (guide).
[INFO DRT-0035] Complete metal3 (guide).
[INFO DRT-0035] Complete via3 (guide).
[INFO DRT-0035] Complete metal4 (guide).
[INFO DRT-0035] Complete via4 (guide).
[INFO DRT-0035] Complete metal5 (guide).
[INFO DRT-0035] Complete via5 (guide).
[INFO DRT-0035] Complete metal6 (guide).
[INFO DRT-0035] Complete via6 (guide).
[INFO DRT-0035] Complete metal7 (guide).
[INFO DRT-0035] Complete via7 (guide).
[INFO DRT-0035] Complete metal8 (guide).
[INFO DRT-0035] Complete via8 (guide).
[INFO DRT-0035] Complete metal9 (guide).
[INFO DRT-0035] Complete via9 (guide).
[INFO DRT-0035] Complete metal10 (guide).
[INFO DRT-0036] active guide region query size = 0.
[INFO DRT-0036] FR_VIA guide region query size = 0.
[INFO DRT-0036] metal1 guide region query size = 758.
[INFO DRT-0036] via1 guide region query size = 0.
[INFO DRT-0036] metal2 guide region query size = 545.
[INFO DRT-0036] via2 guide region query size = 0.
[INFO DRT-0036] metal3 guide region query size = 282.
[INFO DRT-0036] via3 guide region query size = 0.
[INFO DRT-0036] metal4 guide region query size = 32.
[INFO DRT-0036] via4 guide region query size = 0.
[INFO DRT-0036] metal5 guide region query size = 31.
[INFO DRT-0036] via5 guide region query size = 0.
[INFO DRT-0036] metal6 guide region query size = 24.
[INFO DRT-0036] via6 guide region query size = 0.
[INFO DRT-0036] metal7 guide region query size = 0.
[INFO DRT-0036] via7 guide region query size = 0.
[INFO DRT-0036] metal8 guide region query size = 0.
[INFO DRT-0036] via8 guide region query size = 0.
[INFO DRT-0036] metal9 guide region query size = 0.
[INFO DRT-0036] via9 guide region query size = 0.
[INFO DRT-0036] metal10 guide region query size = 0.
[INFO DRT-0179] Init gr pin query.
[INFO DRT-0245] skipped writing guide updates to database.
[INFO DRT-0185] Post process initialize RPin region query.
[INFO DRT-0181] Start track assignment.
[INFO DRT-0184] Done with 601 vertical wires in 1 frboxes and 1071 horizontal wires in 1 frboxes.
[INFO DRT-0186] Done with 119 vertical wires in 1 frboxes and 155 horizontal wires in 1 frboxes.
[INFO DRT-0182] Complete track assignment.
[INFO DRT-0267] cpu time = 00:00:00, elapsed time = 00:00:00, memory = 101.50 (MB), peak = 101.50 (MB)
[INFO DRT-0187] Start routing data preparation.
[INFO DRT-0267] cpu time = 00:00:00, elapsed time = 00:00:00, memory = 101.50 (MB), peak = 101.50 (MB)
[INFO DRT-0194] Start detail routing.
[INFO DRT-0195] Start 0th optimization iteration.
    Completing 10% with 0 violations.
    elapsed time = 00:00:00, memory = 106.90 (MB).
    Completing 20% with 0 violations.
    elapsed time = 00:00:00, memory = 108.62 (MB).
    Completing 30% with 0 violations.
    elapsed time = 00:00:00, memory = 106.33 (MB).
    Completing 40% with 0 violations.
    elapsed time = 00:00:00, memory = 106.33 (MB).
    Completing 50% with 5 violations.
    elapsed time = 00:00:00, memory = 115.41 (MB).
    Completing 60% with 5 violations.
    elapsed time = 00:00:01, memory = 111.01 (MB).
    Completing 70% with 17 violations.
    elapsed time = 00:00:01, memory = 115.13 (MB).
    Completing 80% with 17 violations.
    elapsed time = 00:00:02, memory = 115.13 (MB).
    Completing 90% with 20 violations.
    elapsed time = 00:00:02, memory = 115.16 (MB).
    Completing 100% with 20 violations.
    elapsed time = 00:00:02, memory = 115.16 (MB).
[INFO DRT-0199] Number of violations = 59.
Viol/Layer    metal1 metal2 metal3  via4  via5
Cut Spacing      0      0      0      1      1
Metal Spacing     1      2      0      0      0
Recheck          0     32      7      0      0
Short            1     14      0      0      0
[INFO DRT-0267] cpu time = 00:00:04, elapsed time = 00:00:03, memory = 664.04 (MB), peak = 664.04 (MB)
Total wire length = 1559 um.
Total wire length on LAYER metal1 = 0 um.
Total wire length on LAYER metal2 = 823 um.
Total wire length on LAYER metal3 = 657 um.
Total wire length on LAYER metal4 = 22 um.
Total wire length on LAYER metal5 = 7 um.
Total wire length on LAYER metal6 = 48 um.
Total wire length on LAYER metal7 = 0 um.
Total wire length on LAYER metal8 = 0 um.
Total wire length on LAYER metal9 = 0 um.
Total wire length on LAYER metal10 = 0 um.
Total number of vias = 1637.
Up-via summary (total 1637):.

```

```

-----
active      0
metal1     937
metal2     612
metal3      33
metal4      31
metal5      24
metal6       0
metal7       0
metal8       0
metal9       0
-----
1637

```

```

[INFO DRT-0195] Start 1st optimization iteration.
    Completing 10% with 59 violations.
    elapsed time = 00:00:00, memory = 664.04 (MB).
    Completing 20% with 59 violations.
    elapsed time = 00:00:00, memory = 664.04 (MB).
    Completing 30% with 59 violations.
    elapsed time = 00:00:00, memory = 664.04 (MB).
    Completing 40% with 59 violations.
    elapsed time = 00:00:00, memory = 664.04 (MB).
    Completing 50% with 41 violations.
    elapsed time = 00:00:01, memory = 664.04 (MB).
    Completing 60% with 41 violations.
    elapsed time = 00:00:01, memory = 664.29 (MB).

```

```
Completing 70% with 35 violations.
elapsed time = 00:00:01, memory = 664.29 (MB).
Completing 80% with 35 violations.
elapsed time = 00:00:01, memory = 664.29 (MB).
Completing 90% with 18 violations.
elapsed time = 00:00:02, memory = 665.84 (MB).
Completing 100% with 10 violations.
elapsed time = 00:00:02, memory = 665.84 (MB).
[INFO DRT-0199] Number of violations = 10.
Viol/Layer      metal2
Metal Spacing    3
Short            7
[INFO DRT-0267] cpu time = 00:00:02, elapsed time = 00:00:02, memory = 665.84 (MB), peak = 665.84 (MB)
Total wire length = 1536 um.
Total wire length on LAYER metal1 = 0 um.
Total wire length on LAYER metal2 = 818 um.
Total wire length on LAYER metal3 = 646 um.
Total wire length on LAYER metal4 = 19 um.
Total wire length on LAYER metal5 = 7 um.
Total wire length on LAYER metal6 = 43 um.
Total wire length on LAYER metal7 = 0 um.
Total wire length on LAYER metal8 = 0 um.
Total wire length on LAYER metal9 = 0 um.
Total wire length on LAYER metal10 = 0 um.
Total number of vias = 1626.
Up-via summary (total 1626):.
```

```
-----
active      0
metal1     936
metal2     600
metal3      35
metal4      31
metal5      24
metal6       0
metal7       0
metal8       0
metal9       0
-----
1626
```

```
[INFO DRT-0195] Start 2nd optimization iteration.
Completing 10% with 10 violations.
elapsed time = 00:00:00, memory = 665.84 (MB).
Completing 20% with 10 violations.
elapsed time = 00:00:00, memory = 665.84 (MB).
Completing 30% with 10 violations.
elapsed time = 00:00:00, memory = 665.84 (MB).
Completing 40% with 10 violations.
elapsed time = 00:00:00, memory = 665.84 (MB).
Completing 50% with 10 violations.
elapsed time = 00:00:00, memory = 665.84 (MB).
Completing 60% with 10 violations.
elapsed time = 00:00:00, memory = 665.84 (MB).
Completing 70% with 10 violations.
elapsed time = 00:00:01, memory = 665.84 (MB).
Completing 80% with 10 violations.
elapsed time = 00:00:01, memory = 665.84 (MB).
Completing 90% with 10 violations.
elapsed time = 00:00:01, memory = 665.84 (MB).
Completing 100% with 8 violations.
elapsed time = 00:00:01, memory = 666.61 (MB).
[INFO DRT-0199] Number of violations = 8.
Viol/Layer      metal2
Metal Spacing    2
Short            6
[INFO DRT-0267] cpu time = 00:00:02, elapsed time = 00:00:02, memory = 666.61 (MB), peak = 666.61 (MB)
Total wire length = 1526 um.
Total wire length on LAYER metal1 = 0 um.
Total wire length on LAYER metal2 = 807 um.
Total wire length on LAYER metal3 = 643 um.
Total wire length on LAYER metal4 = 26 um.
Total wire length on LAYER metal5 = 7 um.
Total wire length on LAYER metal6 = 42 um.
Total wire length on LAYER metal7 = 0 um.
Total wire length on LAYER metal8 = 0 um.
Total wire length on LAYER metal9 = 0 um.
Total wire length on LAYER metal10 = 0 um.
Total number of vias = 1618.
Up-via summary (total 1618):.
```

```
-----
active      0
metal1     936
metal2     590
metal3      37
metal4      31
metal5      24
metal6       0
metal7       0
metal8       0
metal9       0
-----
1618
```

```
[INFO DRT-0195] Start 3rd optimization iteration.
Completing 10% with 8 violations.
elapsed time = 00:00:00, memory = 666.61 (MB).
Completing 20% with 8 violations.
elapsed time = 00:00:00, memory = 666.61 (MB).
Completing 30% with 8 violations.
elapsed time = 00:00:00, memory = 666.61 (MB).
Completing 40% with 8 violations.
elapsed time = 00:00:00, memory = 666.61 (MB).
Completing 50% with 3 violations.
elapsed time = 00:00:00, memory = 666.61 (MB).
Completing 60% with 3 violations.
elapsed time = 00:00:00, memory = 671.51 (MB).
Completing 70% with 0 violations.
elapsed time = 00:00:00, memory = 671.51 (MB).
Completing 80% with 0 violations.
elapsed time = 00:00:00, memory = 671.51 (MB).
Completing 90% with 0 violations.
elapsed time = 00:00:00, memory = 671.51 (MB).
Completing 100% with 0 violations.
elapsed time = 00:00:00, memory = 671.51 (MB).
[INFO DRT-0199] Number of violations = 0.
[INFO DRT-0267] cpu time = 00:00:00, elapsed time = 00:00:00, memory = 671.51 (MB), peak = 671.51 (MB)
Total wire length = 1529 um.
Total wire length on LAYER metal1 = 0 um.
Total wire length on LAYER metal2 = 806 um.
```

Total wire length on LAYER metal3 = 647 um.
Total wire length on LAYER metal4 = 25 um.
Total wire length on LAYER metal5 = 7 um.
Total wire length on LAYER metal6 = 42 um.
Total wire length on LAYER metal7 = 0 um.
Total wire length on LAYER metal8 = 0 um.
Total wire length on LAYER metal9 = 0 um.
Total wire length on LAYER metal10 = 0 um.
Total number of vias = 1624.
Up-via summary (total 1624):.

```
-----  
active      0  
metal1     936  
metal2     598  
metal3      35  
metal4      31  
metal5      24  
metal6       0  
metal7       0  
metal8       0  
metal9       0  
-----  
1624
```

[INFO DRT-0198] Complete detail routing.
Total wire length = 1529 um.
Total wire length on LAYER metal1 = 0 um.
Total wire length on LAYER metal2 = 806 um.
Total wire length on LAYER metal3 = 647 um.
Total wire length on LAYER metal4 = 25 um.
Total wire length on LAYER metal5 = 7 um.
Total wire length on LAYER metal6 = 42 um.
Total wire length on LAYER metal7 = 0 um.
Total wire length on LAYER metal8 = 0 um.
Total wire length on LAYER metal9 = 0 um.
Total wire length on LAYER metal10 = 0 um.
Total number of vias = 1624.
Up-via summary (total 1624):.

```
-----  
active      0  
metal1     936  
metal2     598  
metal3      35  
metal4      31  
metal5      24  
metal6       0  
metal7       0  
metal8       0  
metal9       0  
-----  
1624
```

[INFO DRT-0267] cpu time = 00:00:10, elapsed time = 00:00:09, memory = 671.51 (MB), peak = 671.51 (MB)

[INFO DRT-0180] Post processing.
Took 13 seconds: detailed_route -output_drc ./reports/nangate45/UART16/base/5_route_drc.rpt -output_maze ./results/nangate45/UART16/base/maze.log -bottom_routing_layer metal2 -top_routing_layer
[INFO ANT-0002] Found 0 net violations.
[INFO ANT-0001] Found 0 pin violations.
Elapsed time: 0:13.78[h]:min:sec. CPU time: user 17.86 sys 0.58 (133%). Peak memory: 688192KB.
Running fillcell.tcl, stage 5_3_fillcell
[INFO ORD-0030] Using 2 thread(s).
filler_placement "FILLCELL_X1 FILLCELL_X2 FILLCELL_X4 FILLCELL_X8 FILLCELL_X16 FILLCELL_X32"
[INFO DPL-0001] Placed 375 filler instances.
Elapsed time: 0:00.50[h]:min:sec. CPU time: user 0.45 sys 0.05 (99%). Peak memory: 87288KB.
cp ./results/nangate45/UART16/base/5_3_fillcell.odb ./results/nangate45/UART16/base/5_route.odb
cp ./results/nangate45/UART16/base/5_1_grt.sdc ./results/nangate45/UART16/base/5_route.sdc
Running density_fill.tcl, stage 6_1_fill
[INFO ORD-0030] Using 2 thread(s).
exec cp ./results/nangate45/UART16/base/5_route.odb ./results/nangate45/UART16/base/6_1_fill.odb
Elapsed time: 0:00.52[h]:min:sec. CPU time: user 0.46 sys 0.05 (99%). Peak memory: 86376KB.
cp ./results/nangate45/UART16/base/5_route.sdc ./results/nangate45/UART16/base/6_1_fill.sdc
Running final_report.tcl, stage 6_report
[INFO ORD-0030] Using 2 thread(s).
Deleted 0 routing obstructions
[INFO RCX-0431] Defined process_corner X with ext_model_index 0
[INFO RCX-0029] Defined extraction corner X
[INFO RCX-0008] extracting parasitics of UART16 ...
[INFO RCX-0435] Reading extraction model file ./content/OpenROAD-flow-scripts/flow/platforms/nangate45/rcx_patterns.rules ...
[INFO RCX-0436] RC segment generation UART16 (max_merge_res 50.0) ...
[INFO RCX-0040] Final 897 rc segments
[INFO RCX-0439] Coupling Cap extraction UART16 ...
[INFO RCX-0440] Coupling threshold is 0.1000 fF, coupling capacitance less than 0.1000 fF will be grounded.
[INFO RCX-0043] 1530 wires to be extracted
[INFO RCX-0442] 51% completion -- 789 wires have been extracted
[INFO RCX-0442] 100% completion -- 1530 wires have been extracted
[INFO RCX-0045] Extract 412 nets, 1244 rsegs, 1244 caps, 1552 ccs
[INFO RCX-0015] Finished extracting UART16.
[INFO RCX-0016] Writing SPEF ...
[INFO RCX-0443] 412 nets finished
[INFO RCX-0017] Finished writing SPEF ...
[INFO PSM-0022] Using 1.100V for VDD
[WARNING PSM-0016] Voltage pad location (VSRC) file not specified, defaulting pad location to checkerboard pattern on core area.
[WARNING PSM-0017] X direction bump pitch is not specified, defaulting to 140um.
[WARNING PSM-0018] Y direction bump pitch is not specified, defaulting to 140um.
[WARNING PSM-0063] Specified bump pitches of 140.000 and 140.000 are less than core width of 35.340 or core height of 35.000. Changing bump location to the center of the die at (18.810, 18.900)
[WARNING PSM-0065] VSRC location not specified, using default checkerboard pattern with one VDD every size bumps in x-direction and one in two bumps in the y-direction
[INFO PSM-0076] Setting metal node density to be standard cell height times 5.
[WARNING PSM-0030] VSRC location at (18.810um, 18.900um) and size 10.000um, is not located on an existing power stripe node. Moving to closest node at (31.140um, 19.600um).
[INFO PSM-0031] Number of PDN nodes on net VDD = 132.
[INFO PSM-0064] Number of voltage sources = 1.
[INFO PSM-0040] All PDN stripes on net VDD are connected.
IR report #####
Corner: default
Worstcase voltage: 1.09e+00 V
Average IR drop : 3.63e-03 V
Worstcase IR drop: 1.32e-02 V

[INFO PSM-0022] Using 0.000V for VSS
[WARNING PSM-0016] Voltage pad location (VSRC) file not specified, defaulting pad location to checkerboard pattern on core area.
[WARNING PSM-0017] X direction bump pitch is not specified, defaulting to 140um.
[WARNING PSM-0018] Y direction bump pitch is not specified, defaulting to 140um.
[WARNING PSM-0063] Specified bump pitches of 140.000 and 140.000 are less than core width of 35.340 or core height of 35.000. Changing bump location to the center of the die at (18.810, 18.900)
[WARNING PSM-0065] VSRC location not specified, using default checkerboard pattern with one VDD every size bumps in x-direction and one in two bumps in the y-direction
[INFO PSM-0076] Setting metal node density to be standard cell height times 5.
[WARNING PSM-0030] VSRC location at (18.810um, 18.900um) and size 10.000um, is not located on an existing power stripe node. Moving to closest node at (3.140um, 18.200um).
[INFO PSM-0031] Number of PDN nodes on net VSS = 132.
[INFO PSM-0064] Number of voltage sources = 1.
[INFO PSM-0040] All PDN stripes on net VSS are connected.
IR report

```
Corner: default
Worstcase voltage: 1.47e-02 V
Average IR drop : 3.81e-03 V
Worstcase IR drop: 1.47e-02 V
#####
Cell usage report:
  Fill cells: 375
  Tap cells: 50
  Buffer/inverters: 58
  Clock buffer/inverters: 3
  Timing Repair Buffer/inverters: 22
  Sequential cells: 59
  Complex combinational cells: 177
Report metrics stage 6, finish...

=====
finish report_design_area
-----
Design area 542 u^2 44% utilization.
Elapsed time: 0:00.78[h:min:sec. CPU time: user 0.70 sys 0.07 (99%). Peak memory: 114692KB.
cp /content/OpenROAD-flow-scripts/flow/platforms/nangate45/lef/NangateOpenCellLibrary.tech.lef ./objects/nangate45/UART16/base/klayout_tech.lef
SC_LEF_RELATIVE_PATH=$(env('FLOW_HOME'))/platforms/nangate45/lef/NangateOpenCellLibrary.mod.lef"; \
OTHER_LEFS_RELATIVE_PATHS=$(echo "<lef-files>$(realpath --relative-to=./results/nangate45/UART16/base ./objects/nangate45/UART16/base/klayout_tech.lef)</lef-files>"; \
sed 's,<lef-files>.*</lef-files>,<lef-files>' "$SC_LEF_RELATIVE_PATH"</lef-files>"$OTHER_LEFS_RELATIVE_PATHS"',g' /content/OpenROAD-flow-scripts/flow/platforms/nangate45/FreePDK45.lyt > ./obje
sed -i 's,<map-file>.*</map-file>,<map-file>/content/OpenROAD-flow-scripts/flow/platforms/nangate45/*map</map-file>,g' ./objects/nangate45/UART16/base/klayout.lyt
(env time -f 'Elapsed time: %E[h:min:sec. CPU time: user %U sys %S (%P). Peak memory: %MKB.' stdbuf -o L /content/conda-env/bin/klayout -zz -rd design_name=UART16 \
  -rd in_def=./results/nangate45/UART16/base/6_final.def \
  -rd in_files="/content/OpenROAD-flow-scripts/flow/platforms/nangate45/gds/NangateOpenCellLibrary.gds " \
  -rd seal_file="" \
  -rd out_file=./results/nangate45/UART16/base/6_1_merged.gds \
  -rd tech_file=./objects/nangate45/UART16/base/klayout.lyt \
  -rd layer_map= \
  -r /content/OpenROAD-flow-scripts/flow/util/def2stream.py) 2>&1 | tee /content/OpenROAD-flow-scripts/flow/logs/nangate45/UART16/base/6_1_merge.log
[INFO] Reporting cells prior to loading DEF ...
[INFO] Reading DEF ...
[INFO] Clearing cells...
[INFO] Merging GDS/OAS files...
      /content/OpenROAD-flow-scripts/flow/platforms/nangate45/gds/NangateOpenCellLibrary.gds
[INFO] Copying toplevel cell 'UART16'
[INFO] Checking for missing cell from GDS/OAS...
[INFO] Found GDS_ALLOW_EMPTY variable.
[INFO] All LEF cells have matching GDS/OAS cells
[INFO] Checking for orphan cell in the final layout...
[INFO] No orphan cells
[INFO] Writing out GDS/OAS './results/nangate45/UART16/base/6_1_merged.gds'
Elapsed time: 0:00.96[h:min:sec. CPU time: user 0.51 sys 0.09 (62%). Peak memory: 102788KB.
cp results/nangate45/UART16/base/6_1_merged.gds results/nangate45/UART16/base/6_final.gds
echo "Did final file"
Did final file
Done
```